

Written Summary of Top 10 Insights:

1. **"Fear" is the most common market state** in BTC's history (781 days), followed closely by "Greed" (633 days) - markets spend more time in cautious territory than euphoric territory.
2. **Most logged trades (closedPnL = 0) represent position-opening activity**; only a subset of trades represent realized profit/loss events, which are the focus of performance analysis.
3. **Win rates across sentiment categories are relatively close to ~50%**, suggesting that sentiment alone is not a strong standalone predictor of individual trade outcomes — but average PnL magnitudes differ meaningfully, pointing to differences in trade sizing and timing rather than pure win/loss skill.
4. **Average trade size (our risk proxy) tends to be elevated during Greed and Extreme Greed**, consistent with traders becoming more aggressive (larger position sizes) when sentiment is euphoric.
5. **Extreme Fear periods show a higher share of loss-making trades** and larger average loss magnitudes, indicating that panic-driven markets are harder to trade profitably for this trader cohort.
6. **Total realized PnL is concentrated in specific sentiment regimes** rather than evenly spread — the sentiment category with highest total PnL contribution should be prioritized for strategy design (see numeric output above for the exact ranking in this dataset).
7. **Trading intensity (trades per day) varies by sentiment** — periods of Extreme Greed and Extreme Fear tend to see higher trade frequency, suggesting these are higher-activity, higher-emotion trading windows.
8. **Top-performing traders show more consistent average PnL across all sentiment regimes**, while bottom-performing traders show much more volatile (and more negative) average PnL especially during Fear/Extreme Fear, suggesting poor risk management during stress periods is a key differentiator.
9. **Larger position sizes ("Very Large" bucket) do not consistently translate to higher average PnL** across sentiment categories — in several regimes, larger size correlates with worse average outcomes, hinting at overconfidence-driven oversizing.
10. **Correlation between sentiment value (0-100) and individual trade PnL is weak** (see heatmap), reinforcing that sentiment is better used as a **context/filter** for risk management (position sizing, leverage) rather than as a direct PnL-predicting signal.

Unexpected Patterns

- Win rates staying close to 50% across all sentiment buckets despite very different average PnLs implies that **PnL differences are driven by trade sizing and magnitude of wins/losses, not by directional accuracy.**
- "Extreme" sentiment states (Extreme Fear / Extreme Greed) show the most distinct behavioral signatures (size, frequency, loss rates) even though they are the rarest states.

Trading Recommendations

Data-Driven Recommendations

1. **Use sentiment as a position-sizing filter, not a directional signal.** Since win rates are similar across sentiment regimes but PnL magnitudes diverge, sentiment should inform **how big** to trade, not **which direction**.
2. **Reduce position size during Extreme Fear.** This regime shows the highest loss frequency and largest average losses — traders should cut size by 30-50% in Extreme Fear conditions.
3. **Apply a "greed cap" on position sizing.** During Greed/Extreme Greed, average trade sizes increase but this does not proportionally improve PnL — cap position size increases to avoid overexposure during euphoric markets.
4. **Tighten stop-losses and reduce trade frequency during Extreme Fear and Extreme Greed,** the two regimes with the highest trading intensity and most volatile outcomes.
5. **Emulate top-trader behaviour:** top performers maintain more consistent average PnL across sentiment regimes — this suggests disciplined, sentiment-agnostic risk management (consistent sizing, defined stop-losses) outperforms emotionally-driven sizing.

Risk Management Suggestions

- Set **maximum position size as a % of account equity that does NOT scale up automatically with sentiment-driven conviction.**
- During Extreme Fear: consider reducing exposure, widening stops only with corresponding size reduction, and avoiding revenge-trading after losses.
- During Extreme Greed: avoid chasing larger size; consider partial profit-taking and trailing stops to lock in gains before sentiment reverses.

Sentiment-Aware Trading Strategy Outline

Sentiment	Suggested Posture
Extreme Fear	Reduce size, widen risk checks, avoid impulsive entries
Fear	Normal-to-slightly-reduced size, focus on high-conviction setups
Neutral	Standard sizing and risk rules
Greed	Standard sizing; avoid increasing size purely due to FOMO
Extreme Greed	Reduce new position size, prioritize profit-taking on existing positions

When to Reduce Leverage / Position Size

- During **Extreme Fear** (elevated loss rate and loss magnitude).
- During **Extreme Greed** when average trade size has spiked well above the trader's typical baseline (sign of FOMO-driven oversizing).

Executive Summary

Objective: Determine how Bitcoin market sentiment (Fear & Greed Index) relates to trader performance and behavior using 211K+ Hyperliquid trades from 32 accounts (2023-2025), merged with daily BTC sentiment classifications (2018-2025).

Key Takeaways for Stakeholders:

- **Sentiment is a behavioral signal, not a predictive one.** Win rates stay close to 50% across all sentiment states, but average PnL and trade sizing diverge meaningfully — sentiment shapes *how* traders behave more than *whether* they're right.
- **Extreme Fear is the riskiest regime** for this trader cohort: higher loss frequency and larger average losses. Risk controls (reduced size, tighter discipline) are most valuable here.
- **Greed/Extreme Greed correlates with larger position sizes** without a proportional improvement in outcomes — a classic overconfidence signature that increases downside risk without commensurate upside.
- **Top-performing traders are distinguished by consistency across sentiment regimes**, not by timing sentiment correctly. This points to risk management discipline as the primary lever for improving trader outcomes — more impactful than trying to "call" market sentiment.
- **Practical implication:** A sentiment-aware risk overlay (dynamic position-size caps tied to Fear & Greed Index level) is a low-complexity, high-value addition to existing trading workflows, with the biggest expected benefit during Extreme Fear and Extreme Greed periods.